Sandeep Yadav

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SUMMARY

Computer Science undergraduate at NIT Jalandhar (CGPA 7.86/10, top 20% of class) with hands-on full-stack development experience. Independently designed and built a ballistic computation research system addressing a real Indian Army stakeholder need, now backed by an institutional endorsement and an active DRDO procurement proposal. Deployed a production web app on Microsoft Azure and solved 400+ DSA problems on LeetCode. Seeking a Software Engineering internship to build scalable, high-performance systems.

TECHNICAL SKILLS

Programming Languages : Python, C, C++, JavaScript, SQL

Web Technologies : HTML, CSS, JavaScript, DOM Manipulation, Responsive Design, Full Stack Web Development

Backend &APIs : RESTful APIs, Flask, HTTP, JSON, CRUD Operations

Cloud & DevOps: Microsoft Azure, Azure App Service, Git, GitHub, Version Control

CS Fundamentals: DSA, Object-Oriented Programming (OOP), Computer Networks, Operating Systems, Machine Learning

Soft Skills : Communication , Agile Development, Problem Solving, Teamwork & Collaboration, Adaptability, Time Management

RESEARCH EXPERIENCE

Ballistic Computation System — Research Project ***(Demo video)***
Endorsed by NIT Jalandhar • DRDO procurement proposal in progress ***Dec 2025 – July 2026***

- Identified an operational problem raised by an Indian Army stakeholder and independently designed a full-stack ballistic computation engine, including the core computation logic and a real-time dashboard for non-technical personnel to operate directly.
- Early internal testing indicates calculation turnaround dropping from roughly 12 minutes (manual) to about 2 minutes — an ~80% reduction for representative cases; formal validation is ongoing.
- Migrating the application to a Tauri desktop app with a Rust backend to strengthen security through OS-level controls, reducing the browser-based attack surface; testing is in progress.
- Independently handled the full project lifecycle — requirements gathering with the stakeholder, design, development, and testing — and am now working with my institution toward a formal DRDO evaluation and procurement pathway.

PROJECTS

AgriMarketProfitOptimiser ***Academic Project | NIT Jalandhar***
Python, Data Analysis, Optimisation *August 2024 – December 2024*

- Built a Python decision-support tool to identify the most profitable markets for selling crops.
- Analysed multi-market crop pricing data using Pandas and NumPy, and implemented an optimisation algorithm to recommend optimal selling locations.
- Replaced manual market comparison with a fully data-driven recommendation workflow.

Mess Activity Tracker ***Personal Project***
Python, REST API, iPhone Shortcuts, Geofencing *June 2025 – July 2025*

- Built an automated mess entry/exit logging system using iPhone Shortcuts geofencing triggers, removing the need for manual daily tracking.
- Developed a Python REST API backend handling real-time POST requests with sub-500ms response times.
- Persisted geofence event logs in a MySQL storage layer for reliable record-keeping.

Research Document Dashboard ***Academic Project | NIT Jalandhar***
JavaScript, HTML, CSS, Microsoft Azure App Service *August 2025 – December 2025*

- Independently deployed a full-stack web dashboard on Microsoft Azure, serving 50+ academic documents.
- Implemented dynamic client-side filtering that cut document retrieval time by roughly 70%.
- Took the project from local development to a live Azure deployment in a single day, with a responsive UI for both desktop and mobile.

EDUCATION

B.Tech — Computer Science & Engineering **August 2024 – May 2028**
NIT Jalandhar (Dr. B.R. Ambedkar National Institute of Technology), Punjab, India ***CGPA: 7.86 / 10 (GPA : 3.14/4)***

- Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems | Top 20% of class by CGPA.

CERTIFICATIONS

- Microsoft Azure — Hands-on cloud deployment via Azure App Service | December 2025
- LeetCode — 400+ problems solved: Arrays, Linked Lists, Trees, Dynamic Programming | 2024 – Present
- Codeforces — Rated competitive programmer, 5+ contests completed (Handle: sandeepy.cs.24) | 2024 – Present
- Participated in 1-2 Hackathons — achieving finalist/winner position in competitive coding and development challenges
- Mobile Security Bootcamp Certificate — IIT Jammu and NIT Jalandhar collaboration | 2024

ACHIEVEMENTS

- Completed and delivered 1 classified Indian Army defence application as sole developer with 100% accuracy (December 2025 – Ongoing)
- Solved 400+ LeetCode problems across Arrays, Trees, DP and Graph categories; competed in 5+ Codeforces rated contests
- Achieved 2nd Place in Mobile Security Bootcamp conducted in collaboration with IIT Jammu and NIT Jalandhar